

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Experienced and performance-driven ML Operations Engineer with over 9 years of expertise in developing, deploying, and maintaining scalable ML systems and cloud-native services. Adept in MLOps tooling, container orchestration, cloud infrastructure, and production-grade AI deployment pipelines. Passionate about enabling high-performance engineering teams through automation, monitoring, and cross-functional collaboration. Proven track record in fast-paced remote environments and delivering ML solutions that drive business outcomes.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (S3, RDS, SageMaker, EKS, CloudFormation), GCP (Vertex AI, GKE, Cloud Run)
- **MLOps & Orchestration:** MLflow, Kubeflow, Airflow, Weights & Biases, Kafka, Docker, Kubernetes
- **Monitoring & APIs:** Prometheus, Grafana, FastAPI, OpenAPI, GitOps, CI/CD (GitHub Actions, CircleCI)
- **Programming:** Python (PyTorch, TensorFlow, scikit-learn, LangChain), Java, Bash
- **Databases & Search:** PostgreSQL, MySQL, Redis, Elasticsearch, MongoDB, DynamoDB
- **Others:** Terraform, Helm, Jupyter, HuggingFace, REST, Agile/Scrum, NoSQL, SQL

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and deployed end-to-end MLOps pipelines using Kubeflow and MLflow, enabling seamless model training, versioning, and real-time deployment across multiple environments.
- Architected a scalable AI platform leveraging AWS SageMaker, S3, and RDS, accelerating ML lifecycle and reducing manual deployment errors by over 60%.
- Integrated Prometheus and Grafana for full-stack observability and live KPI dashboards to monitor latency, inference accuracy, and GPU resource usage.
- Led the migration of legacy Python models to containerized microservices orchestrated via Kubernetes, reducing deployment time from days to under an hour.
- Developed CI/CD pipelines using GitHub Actions and Terraform to automate infrastructure provisioning and streamline multi-stage testing and deployment.

- Built internal tooling with FastAPI and LangChain for orchestrating vector-based RAG pipelines across enterprise knowledge bases.
- Collaborated with product managers and data scientists to drive Agile-based experimentation cycles, ensuring frequent model releases with robust rollback strategies.
- Implemented Redis and Elasticsearch-based feature stores and similarity search systems, drastically improving retrieval speed and user targeting precision.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Managed the ML inference pipeline for a personalization engine using scikit-learn and TensorFlow, serving models in production via FastAPI and Dockerized services.
- Containerized and deployed ML services to AWS ECS and Kubernetes clusters, utilizing CloudWatch and Grafana for resource and service health monitoring.
- Developed Airflow DAGs for automating data ingestion, preprocessing, model retraining, and deployment, shortening ML cycle duration by 40%.
- Integrated feature flags and A/B testing frameworks for live experimentation, enabling safe deployments and controlled rollout of ML models.
- Worked cross-functionally with frontend and data teams to implement RESTful APIs that exposed ML services and tracked prediction performance over time.
- Created automated rollback strategies in CI/CD pipelines using GitOps and conditional deployment stages to handle failed releases gracefully.
- Maintained multiple databases including PostgreSQL and MongoDB for structured and unstructured data supporting NLP and recommender systems.
- Assisted in evaluating third-party ML libraries and MLOps vendors, leading to a 20% infrastructure cost reduction and improved model traceability.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Developed and optimized ML microservices using PyTorch and Docker, including training jobs for document classification and customer segmentation models.
- Designed event-driven architecture using Kafka for real-time processing of telemetry and behavioral analytics data.
- Wrote internal Python libraries to standardize experiment tracking, model evaluation, and metric logging across multiple ML teams.
- Set up Kubernetes clusters on GCP with GPU node pools to support high-performance training and inference jobs.
- Orchestrated batch model scoring jobs with Airflow and managed resource queues via Celery and RabbitMQ.
- Led a team-wide transition to Git-based CI/CD with GitLab, including container scanning and infrastructure validation workflows.
- Built Grafana dashboards integrated with Prometheus to provide real-time metrics for model drift detection and alerting systems.
- Enhanced the developer experience by maintaining a curated collection of Jupyter templates and API stubs for prototyping AI services.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Collaborated on the implementation of ML-based fraud detection services integrated with client-facing financial applications.
- Assisted in designing early-stage pipeline automation for training logistic regression and decision tree models on transactional datasets.
- Created a centralized configuration management system to standardize ML model parameters and training configurations.
- Wrote Python and Bash scripts to preprocess CSV and JSON datasets, ensuring compatibility with model training formats.
- Conducted cost-analysis experiments comparing SageMaker and GCP Vertex AI for model training scalability and infrastructure resilience.
- Set up lightweight container orchestration using Docker Compose for local ML development workflows.
- Participated in daily stand-ups and sprint retrospectives, contributing to continuous delivery improvements and engineering best practices.
- Authored technical documentation and architecture diagrams for MLDLC workflows to support onboarding and code handover.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Built utility scripts for automated dataset labeling and integrity checks across structured and image-based data sources.
- Supported the migration of models from R scripts to Python, ensuring reproducibility through conda environments and requirements pinning.
- Participated in team code reviews and contributed to the early design of REST APIs for ML model exposure and integration.
- Implemented simple visualization tools using Matplotlib and Seaborn to help data scientists understand model behavior and feature importances.
- Assisted with version control management, Git workflows, and CI pipeline triggers based on tagged model releases.
- Wrote reusable modules for data cleaning and tokenization to support downstream NLP tasks.
- Deployed basic models on virtual machines for evaluation and manually monitored logs for prediction success/failure analysis.
- Documented all internship contributions, presented findings to the team, and helped onboard new interns in a structured way.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- **AWS Certified Machine Learning – Specialty**
 - **Google Professional Machine Learning Engineer**
 - **MLOps Specialization (Coursera - DeepLearning.AI)**
 - **Kubernetes for Machine Learning Engineers (Udacity Nanodegree)**
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PROJECTS

- **Smart Model Deployment Framework**
Built an internal ML Ops platform using FastAPI, Airflow, MLflow, and Kubernetes to deploy and monitor ML models across staging and production environments for a SaaS platform.
- **Scalable RAG Platform**
Designed and implemented a scalable retrieval-augmented generation (RAG) system using LangChain, Redis, and HuggingFace Transformers to power a contextual Q&A chatbot.
- **AI Monitoring Suite**
Developed a monitoring toolkit with Prometheus, Grafana, and custom FastAPI endpoints to track live ML model performance and detect real-time drift and anomalies.